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ColorAptitude — A Digital Platform for Professional Colour Competency Assessment

A three-layer cognitive framework with OKLCH-based stimulus generation and differentiated per-layer validation

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1. Abstract

This document describes a computer-implemented digital platform for professional colour competency assessment, training, and certification. It is positioned as an occupational-qualification instrument in the sense of ASTM E1499-16, distinct from clinical screening or grading tests such as the FM100, Cambridge Colour Test, or Colour Assessment and Diagnosis test.

The platform is organised around a three-layer cognitive framework decomposing professional colour competency into discrimination (detecting that a colour difference exists), attribution (identifying which perceptual dimension has changed), and communication (translating the percept into standardised professional terminology). The three layers are measured by dedicated instruments, scored through independent pipelines, and reported as separate values within a 3×3 competency matrix combining the three layers with hue, lightness, and chroma.

Stimulus generation uses OKLCH with per-hue gamut-aware trial construction. A two-path display calibration architecture distinguishes observer-managed deployment from ASTM E1499-16 §5.3 compliant qualification, with unified scoring and certificate-tier annotation. Validation follows a differentiated three-track strategy reflecting the distinct psychometric development stages of the three layers. The temporal architecture of assessment, training, and re-test is grounded in the consolidation kinetics of perceptual learning.

The subject matter disclosed in Sections 3.1 through 3.9 comprises: the three-layer framework, the 3×3 matrix reporting architecture, the differentiated validation strategy, a five-segment Hue Ordering Diagnostic with three red-green opponent segments, the two-path calibration architecture, cardinal-nearest hue classification, OKLCH gamut-aware trial generation, Kuehni-anchored closed-set attribution scoring, and a temporal architecture for assessment, training, and re-test grounded in perceptual-learning consolidation kinetics.

2. Technical Background

This section establishes the minimum context required to identify the innovations of Section 3 as prior art.

2.1 Professional colour competency as a three-part capability

Professional colour judgment in industrial contexts requires three empirically distinguishable capabilities. Discrimination is the perceptual ability to detect that a colour difference exists. Attribution is the cognitive-analytical ability to identify which perceptual dimension has changed and in which direction; neuropsychological double-dissociation evidence and pre-verbal infant categorical perception establish this layer as separable from discrimination at the neural level. Communication is the linguistic-professional ability to translate perceived differences into standardised terminology and to reconstruct a percept from a description; cross-cultural colour-naming research establishes this layer as separable from both lower layers.

Existing clinical instruments (FM100, Cambridge Colour Test, Colour Assessment and Diagnosis test) measure discrimination only. No commercially available instrument known to the author combines all three capabilities in an integrated assessment platform before this publication.

2.2 Colour dimensions and perceptually uniform colour spaces

Three dimensions are inherent in any cylindrical perceptual colour space: hue, lightness, and chroma. CIELAB is approximately uniform but exhibits hue non-uniformities. OKLCH (Ottosson, 2020) provides perceptual uniformity sufficient for single-dimension trial generation. OKLCH itself is prior art of Ottosson and is not claimed.

2.3 ASTM E1499-16 as a qualification framework

ASTM E1499-16 specifies observer-qualification requirements across selection, baseline evaluation, training, formal qualification, retraining, and re-qualification. The standard is prior art and is not claimed; implementation of its requirements in a specific integrated digital platform combined with the methods of Section 3 is the subject of this publication.

3. Disclosed Innovations

Each of the nine innovations below is disclosed at concept level for the purpose of establishing prior art. The architectural principle is the disclosed subject matter; specific parameter values, algorithmic implementations, source code, datasets, and operational details are not part of this disclosure and remain the exclusive property of the author and Xconea.

3.1 Three-Layer Competency Framework for Colour Assessment

Background

Existing digital and physical colour-vision instruments measure discrimination only. No existing architecture distinguishes between perceptual discrimination, analytical attribution, and linguistic communication as separable and separately scored components of professional colour competency. The Japanese Colour Aptitude Test (Hirschler, Gay & Ferreira de Oliveira, 2002) includes a Colour-Attribute Discrimination Test as a physical-artefact precedent for attribution measurement, but it is not integrated with discrimination or communication measurement, is not digital, and does not produce dimension-resolved or layer-resolved scoring. Neuropsychological evidence for the separation of attribution from naming (Siuda-Krzywicka et al., 2019; Skelton et al., 2017) establishes the empirical legitimacy of the three-layer distinction but has not been operationalised as an assessment architecture.

Disclosed innovation

The architectural separation of three capabilities — discrimination, attribution, and communication — into independently measured, scored, and reported components within a single integrated assessment platform. Each layer has its own stimulus structure, task demand, scoring pipeline, and reporting output. No aggregation across layers replaces the layer-resolved report. Layer-differentiated training selection follows from the layer-resolved report.

3.2 Three-by-Three Competency Matrix as Reporting Architecture

Background

Existing instruments report a single global score (FM100 Total Error Score) or a few confusion-axis indices (Cambridge protan/deutan/tritan). Neither distinguishes which perceptual dimension a deficit concentrates in nor which cognitive layer it operates at.

Disclosed innovation

A reporting architecture in which three competency layers (discrimination, attribution, communication) are crossed with three perceptual dimensions (hue, lightness, chroma) to produce a 3×3 matrix of nine independently scored cells. Each cell is preserved through the reporting pipeline. Certification is performed at cell level, supporting role-specific competency profiles. Training selection and progress tracking are performed at cell level.

3.3 Differentiated Three-Track Validation Strategy

Background

Conventional psychometric validation is treated as a single activity assuming all measured capabilities sit at comparable development stages. The three-layer framework violates these assumptions: discrimination has extensive normative tradition; attribution has no established normative population; communication has no pre-existing population at the scale required for norm-referenced validation.

Disclosed innovation

A validation strategy using three parallel tracks, one per layer, each matched to that layer's psychometric development stage. Track 1 applies norm-referenced validation to discrimination. Track 2 applies known-groups validation to attribution. Track 3 applies longitudinal construct-validity accumulation to communication. Tier-labelled cell scores allow all three layers to contribute to the competency profile at each validation stage without overreaching the available evidence for any layer. The architectural commitment is the alignment of validation method to development stage.

3.4 Five-Segment Hue Ordering Diagnostic with Three Red-Green Opponent Segments

Background

The Farnsworth-Munsell 100 Hue Test organises 85 movable caps across four trays whose boundaries lie at the four unique hues. Two of these four segments cross red-green opponent transitions. A diagnostic providing additional independent observations of the red-green axis would increase diagnostic power. Fairchild (2018) argued that discrimination tasks align more closely with the five Munsell principal hues than with the four unique hues.

Disclosed innovation

A hue-ordering diagnostic organising the hue circle into five segments at the five Munsell principal hues, of which three segments cross red-green opponent transitions and two do not. The three opponent-crossing segments provide three independent observations of the red-green axis per administration, increasing diagnostic confidence relative to the four-segment FM100 structure. Total cap count and per-segment cap count are configured to preserve administrative comparability with the FM100.

3.5 Two-Path Display Calibration Architecture

Background

Digital colour-assessment instruments on consumer displays face a methodological dilemma. Hardware calibration with spectrophotometers or colorimeters supports formal ASTM E1499-16 §5.3 qualification but is unavailable to most observers. Software-only calibration produces acceptable approximations for training but does not meet §5.3.

Disclosed innovation

A calibration architecture implementing two parallel paths producing identically structured scoring output, with the active path recorded as session metadata and propagated through certification. Path A is a software-indicative wizard suitable for formative use, with output explicitly labelled as not qualifying under §5.3. Path B is a hardware-compliant procedure using a colorimeter and ambient sensor, suitable for formal qualification. Path B additionally implements ISO 3664:2009 P2 viewing-condition compliance logging and mixed-chromatic-adaptation risk logging. Certification tiers reflect the active path; the two-path distinction is visible in the certification record.

3.6 Cardinal-Nearest Hue Classification for Attribution Scoring

Background

Hue-attribution scoring requires a reference partition of the hue circle. Free-naming suffers the circularity formalised by Logvinenko (2012): the observer's naming becomes both measurement and reference. Observer-derived unique-hue calibration reintroduces the circularity at the calibration stage.

Disclosed innovation

A scoring method using a fixed observer-independent reference partition derived from empirical population data on cardinal-hue selections (Kuehni, Hinks, & Shamey, 2008). For each attribution trial, the stimulus hue angle is classified to the nearest population-derived cardinal. The observer selects from a closed set of named hue regions corresponding to the cardinal sectors. The classification is cardinal-nearest at both stimulus and response side; the observer is never asked to free-name and never to select a unique hue as calibration anchor. Population-level bias becomes a measurable property of the observer rather than an invisible offset absorbed into self-referential scoring.

3.7 OKLCH Gamut-Aware Per-Hue Trial Generation

Background

Single-dimension stimulus variation in OKLCH is arithmetically straightforward, but the reproducible gamut of a display does not fill the OKLCH cylinder uniformly. Some hue-lightness combinations admit chroma near full saturation; others admit only reduced chroma before clipping. Clipping introduces cross-dimensional variation, compromising the intended dimensional isolation.

Disclosed innovation

A stimulus-generation method computing per-hue maximum reproducible chroma across the full hue-angle set in a trial set, then applying a uniform chroma value bounded by the minimum-across-hues of these maxima with a calibrated safety margin to accommodate floating-point and rendering variation. The maximum reproducible chroma is computed per hue angle through iterative testing of in-gamut membership; the applied chroma for the trial set is the minimum of these per-hue maxima reduced by the safety margin. The minimum-across-hues selection guarantees that every hue angle in the set can be rendered without clipping, preserving dimensional isolation across the trial set. The method extends symmetrically to lightness-variation and chroma-variation trial sets.

3.8 Kuehni-Anchored Closed-Set Attribution Scoring

Background

Attribution and communication scoring requires correctness determination. Closed-set scoring eliminates interpretive ambiguity at scoring and supports chance-corrected accuracy with known baselines. When the closed set is anchored to population-derived reference data, scoring is methodologically clean and psychometrically defensible.

Disclosed innovation

A scoring architecture using closed response sets for all attribution and communication tasks, anchored to empirical population-derived references (Kuehni, Hinks, & Shamey, 2008) for hue-region tasks, with chance-corrected scoring accounting for response-set cardinality. Hue-region identification, dimension identification, joint dimension-and-magnitude attribution, and direction-and-magnitude descriptor selection each use a calibrated closed set with the corresponding chance baseline. Scoring outputs are directly comparable across task types within each layer and across layers within the matrix.

3.9 Temporal Architecture for Assessment, Training, and Re-Test

Background

Digital platforms integrating measurement, training, and re-test must determine timing relationships between these activities. Karni and Sagi (1993) established that perceptual learning has two temporally distinct components — a fast within-session phase and a slow between-sessions phase requiring a latent consolidation window — and that consolidated gains are retained without further training. Existing instruments do not embody this structure; clinical tests treat each administration as temporally independent. A platform explicitly grounding inter-session interval, training-to-retest cooldown, retention-probe interval, and within-session protocol structure in the consolidation kinetics of perceptual learning is, to the author's knowledge, not present in the prior art.

Disclosed innovation

A temporal architecture specifying scheduling rules grounded in the consolidation kinetics of perceptual learning. The architecture comprises four rule classes: an inter-session interval bounded below by a minimum placing the second session beyond the slow consolidation window; a training-to-retest cooldown ensuring training-induced gain has fully consolidated before measurement; a retention-probe interval evaluating persistence of training-induced gains at a calibrated post-consolidation point; and a within-session warm-up structure mitigating contamination by the fast within-session learning component. The architecture binds assessment, training, and re-test components into a temporally coherent whole rather than treating them as independent operations. Scheduling is enforced through platform-level logic. The architecture extends beyond colour-competency assessment to any platform integrating measurement with perceptual or cognitive training.

4. Cross-Innovation Architecture

4.1 Dependency relations

The three-layer competency framework (§3.1) is the architecture root, defining what is measured. The 3×3 matrix (§3.2) depends on §3.1 and on §3.7 via per-instrument dimension mapping. The differentiated three-track validation strategy (§3.3) depends on §3.1 and is instrument-independent. The Hue Ordering Diagnostic (§3.4) is a discrimination-layer instrument and depends on §3.7 for stimulus construction. The two-path calibration architecture (§3.5) is layer-independent and is a precondition for valid measurement across all instruments. Cardinal-nearest hue classification (§3.6) scores attribution on the hue dimension and depends on §3.8 for cardinal-anchor values. OKLCH gamut-aware trial generation (§3.7) is a stimulus-construction primitive used by §3.4, §3.6, and the lightness/chroma instruments. Kuehni-anchored attribution scoring (§3.8) provides anchor values consumed by §3.6. The temporal architecture (§3.9) operates orthogonally to the eight measurement-and-reporting innovations and constrains all instruments uniformly at platform level.

4.2 Composite architecture

Each of the nine innovations is individually disclosed and individually identifiable. The composite — three-layer measurement, 3×3 matrix reporting, differentiated per-layer validation, two-path calibration, gamut-aware stimulus generation, cardinal-nearest classification, Kuehni-anchored scoring, and consolidation-grounded temporal architecture integrated in a single deployable platform — is a distinct subject of disclosure. No known published assessment architecture combines these elements in an integrated digital platform before this publication.

5. Legal Framework and Publication Record

5.1 Intellectual property ownership

The methods, architectures, and assessment frameworks disclosed are developed by Markus Kotterink (ORCID: 0009-0001-4225-5891). Intellectual property is owned personally; management, licensing, and commercial exploitation are conducted through Xconea — BNK Foundation, which holds an exclusive management mandate. Stichting Nederlands Kleur Instituut (SNKI) operates as a sublicensee of Xconea for specific commercial applications.

5.2 Nature and purpose of this publication

This document is a defensive publication. Its purpose is establishment of public prior art for the subject matter disclosed in Sections 3.1 through 3.9. By publishing a dated, attributed, and identifying description of these innovations, the document places the described subject matter into the public domain with respect to novelty considerations in patent examination worldwide. Under 35 U.S.C. § 102(a)(1) and corresponding provisions in the European Patent Convention (Article 54), the EUTM Regulation, and comparable instruments in other jurisdictions, publicly available descriptions with an established publication date constitute prior art against subsequent patent claims by third parties.

5.3 Reservation of rights

This publication establishes prior art on the architectural concepts disclosed in Sections 3.1 through 3.9. It does not constitute a licence, an authorisation, or an invitation to implement. The implementation of the disclosed architecture — including but not limited to source code, user-interface designs, normative datasets, parameter calibrations, server infrastructure, and brand marks (ColorAptitude, Xconea, SNKI, Nederlandse Kleurenschool, ColorExpertsHub) — remains the exclusive property of the author and Xconea. The author and Xconea reserve all rights in respect of the implementation that are not extinguished by this prior-art disclosure.

5.4 Copyright and licensing

The described technical methods are not themselves copyrightable; the prior-art status established by this publication is governed by the patent-law provisions of the relevant jurisdictions.

5.5 Public accessibility

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5.6 Version and integrity

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Changelog v1.1 → v1.2: Reduction of disclosure to architectural-concept level; removal of implementation parameters, algorithmic details, and operational specifics. The nine innovations of v1.1 (Sections 3.1 through 3.9) and the prior-art dates established by v1.0 and v1.1 are unchanged. This v1.2 document adds no new subject matter; it restricts the scope of disclosure of existing subject matter.

5.7 Citation

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5.8 Warranties and contact

This document is published without warranties of any kind. The author and Xconea make no representations about the validity, scope, or enforceability of any patent application, issued patent, or other intellectual property right, whether their own or of third parties, in connection with the subject matter disclosed. Contact: m.kotterink@kleurinstituut.nl.

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